UNIVERSALITY OF A FIRST ORDER THICKENING OF ALGEBRAIC SPACES IS TESTED ON SCHEMES AND ON AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2963-2977 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2979-3001, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings of algebraic spaces and the lifting condition defining them; the reductions are retained.

Source reference. Original lemma equation-universal-first-order-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2944–2952. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-definition-presentation: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3001–3001. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $h : Z \to X$ be a morphism of algebraic spaces over S. Let $Z \subset Z'$ be a first order thickening over X. The following are equivalent*

1. *$Z \subset Z'$ is a universal first order thickening,*
2. *for any diagram (equation-universal-first-order-thickening) with T' a scheme a unique dotted arrow exists making the diagram commute, and*
3. *for any diagram (equation-universal-first-order-thickening) with T' an affine scheme a unique dotted arrow exists making the diagram commute.*

Proof. The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) are formal. Assume (3) a assume given an arbitrary diagram (equation-universal-first-order-thickening). Choose a presentation $T' = U'/R'$, see Spaces, Definition spaces-definition-presentation. We may assume that $U' = \coprod U'_i$ is a disjoint union of affines, so $R' = U' \times_{T'} U' = \coprod_{i,j} U'_i \times'_T U'_j$. For each pair (i, j) choose an affine open covering $U'_i \times'_T U'_j = \bigcup_k R'_{ijk}$. Denote U_i, R_{ijk} the fibre products with T over T'. Then each $U_i \subset U'_i$ and $R_{ijk} \subset R'_{ijk}$ is a first order thickening of affine schemes. Denote $a_i : U_i \to Z$, resp. $a_{ijk} : R_{ijk} \to Z$

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the composition of $a : T \to Z$ with the morphism $U_i \to T$, resp. $R_{ijk} \to T$. By (3) applied to $a_i : U_i \to Z$ we obtain unique morphisms $a'_i : U'_i \to Z'$. By (3) applied to a_{ijk} we see that the two compositions $R'_{ijk} \to R'_i \to Z'$ and $R'_{ijk} \to R'_j \to Z'$ are equal. Hence $a' = \coprod a'_i : U' = \coprod U'_i \to Z'$ descends to the quotient sheaf $T' = U'/R'$ and we win. $\square$